

Predicting Electricity Power Consumption for Tetuoan city in Zone 1

Final Project Report

STAT 385

May 1, 2026

Team Members and Contributions

Team member	Contribution
Mahi Patel	Presentation Slide, Final report and doing analysis for zone 2 and zone 3(future works)
Rahee Chauhan	EDA, Data cleaning
Mustafa khan	Model building (linear regression, LASSO regression, decision tree)
Usman Tahir	Model evaluation, comparison

Electricity demand and supply management is one of the most important problems in modern energy systems. Accurately predicting power output generated within a specific zone helps grid operators plan resources, reduce waste and avoid blackouts. In this project we focus on zone 1 power output using a dataset that contains weather related and time-based features that are recorded across multiple zones in power distribution network. We chose to focus on zone 1 specifically because it was the most complete and informative zone in the dataset. We also thought that analyzing one zone carefully gives us cleaner and more interpretable results than trying to predict all zones simultaneously. Future work could extend the models to other zones.

Research questions

- Can we predict power output in Zone 1 using weather and environmental features?
- Which variables (like temperature, humidity, wind speed or diffuse flows) are most strongly related to Zone 1 power output?
- Which statistical or machine learning model provides the best prediction performance for this task?
- Do nonlinear models like decision trees outperform simpler models like MLR for this dataset?

To answer these questions, we tried applying three different modeling approaches: Multiple Linear Regression, LASSO Regression and Decision Tree. We evaluated all models on the same training and testing splits to make fair comparison.

2) EXPLORATORY DATA ANALYSIS (EDA)

Before building any models, we explored the dataset carefully to understand its structure, identify patterns and decide which variables to include as predictors. This step helped us with modeling decisions.

Dataset Overview

The dataset contains data collected every 10 mins. It consists of energy related and weather variables from multiple zones. Each row represents one hour of data. Key variables include:

- Temperature(C): Ambient temperature in the zone
- Humidity (%): Relative humidity
- Wind Speed (km/h): Wind speed at the time of measurement
- General Diffuse Flows (GDF): Solar Diffuse irradiance
- Diffuse Flows (DF): Direct Diffuse irradiance
- Date and Time: Timestamp of the observation
- Zone 1 Power Consumption (MW): Target variable

We performed data cleaning by removing missing values, checking for outliers, and verifying that variable types are correct. The data-time variable extracted hour of day, day of week and month. These were added as additional predictors to capture time-based patterns in power usage.

Summary statistics

The table below shows summary statistics for main variables. Zone 1 power output ranged from roughly 13896 to 52204 megawatts with a mean near 32346 megawatts. This indicates meaningful variation that our models need to capture

Variable	Min	Mean	Max
Zone 1 Power (MW)	13,896	32,346	52,204
Temperature(C)	3.247	18.811	40.010
Humidity (%)	11.34	68.26	94.80
Wind Speed(km/h)	0.050	1.960	6.483
General Diffuse Flows	0.004	182.718	1163.000
Diffuse Flows	0.011	75.037	936.000

Visualizations and key observations

We created several plots to understand how each predictor relates to zone 1 power output. Key findings from the EDA are as follows:

Histograms: distribution of Variables

The histograms show the distribution of Temperature, Humidity, Wind Speed and Zone 1 Power Consumption. We can see that temperature is roughly bell shaped centered around 18-22 C. Humidity is skewed toward higher values. Wind speed is heavily concentrated near 0 with a secondary cluster around 5 km/h. Zone 1 power shows a roughly symmetric distribution centered around 30,000- 35,000 MW.

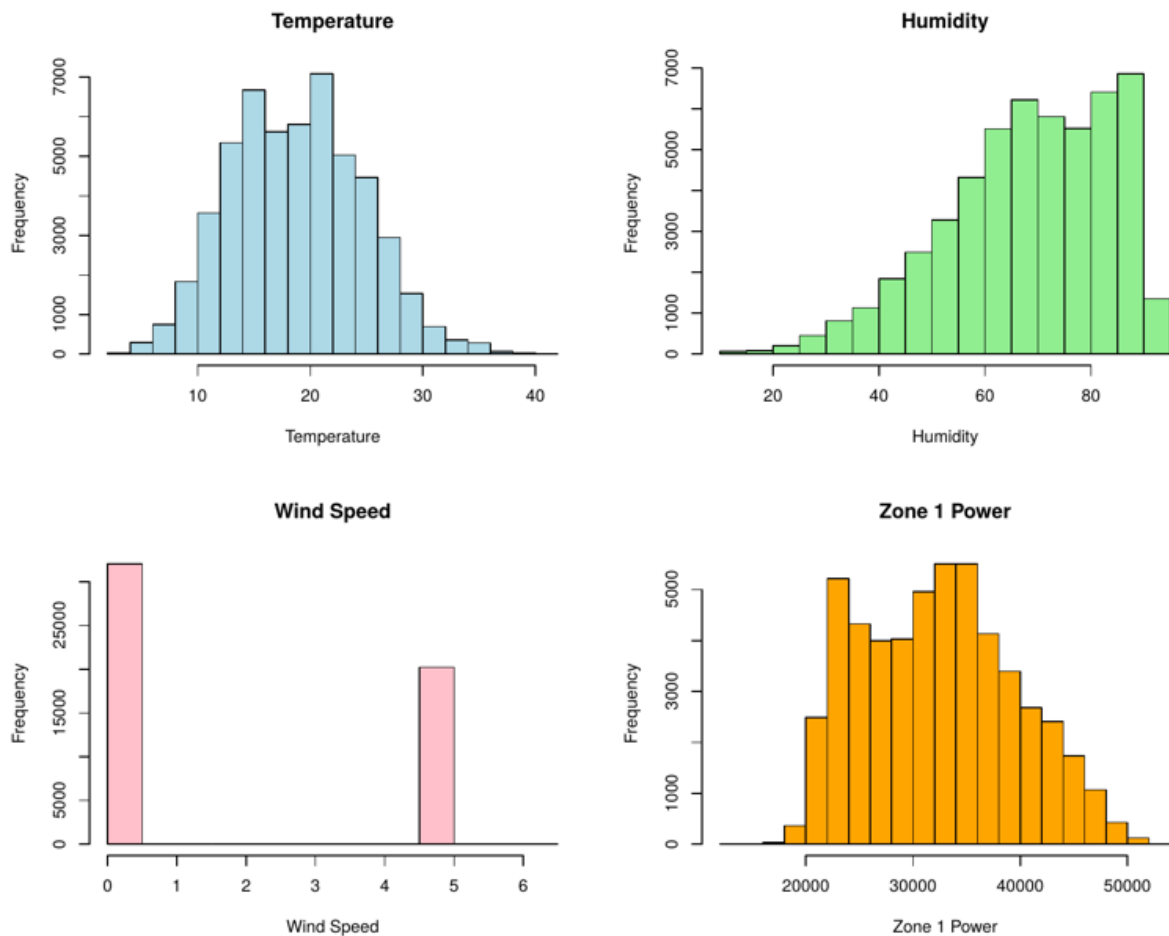
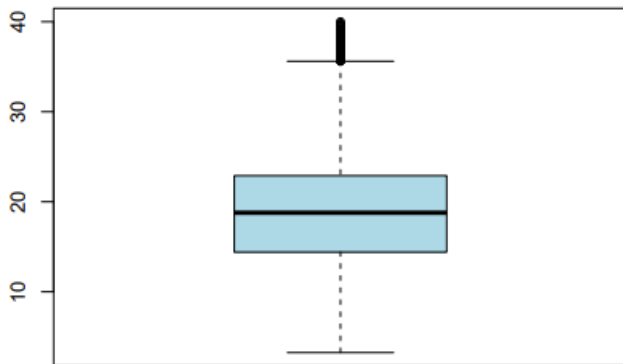


Figure 1: Histograms of Temperature, Humidity, Wind Speed, and Zone 1 Power Consumption

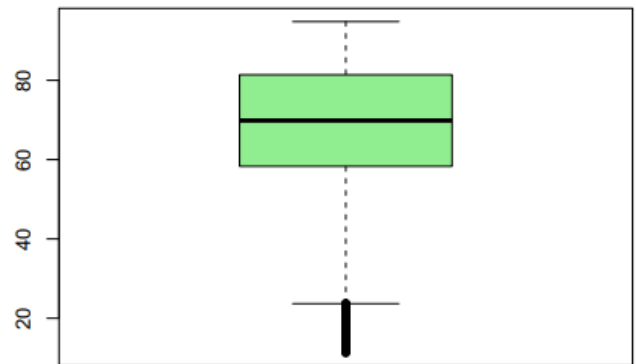
Boxplots

The boxplots help confirm the distribution seen in histogram. The Zone 1 Power boxplot shows an range of roughly 26,000-38,000 MW with median around 32,000 MW.

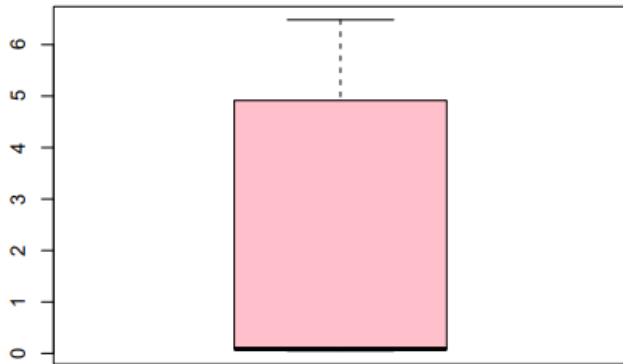
Boxplot: Temperature



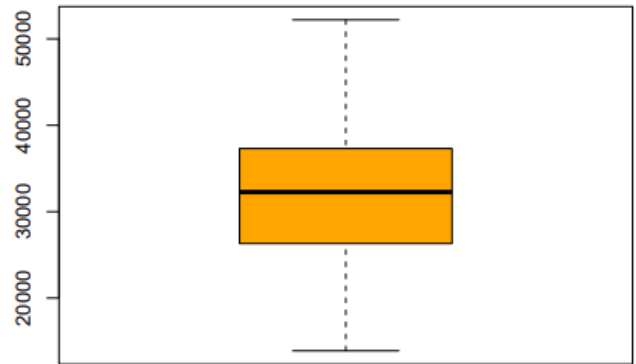
Boxplot: Humidity



Boxplot: Wind Speed

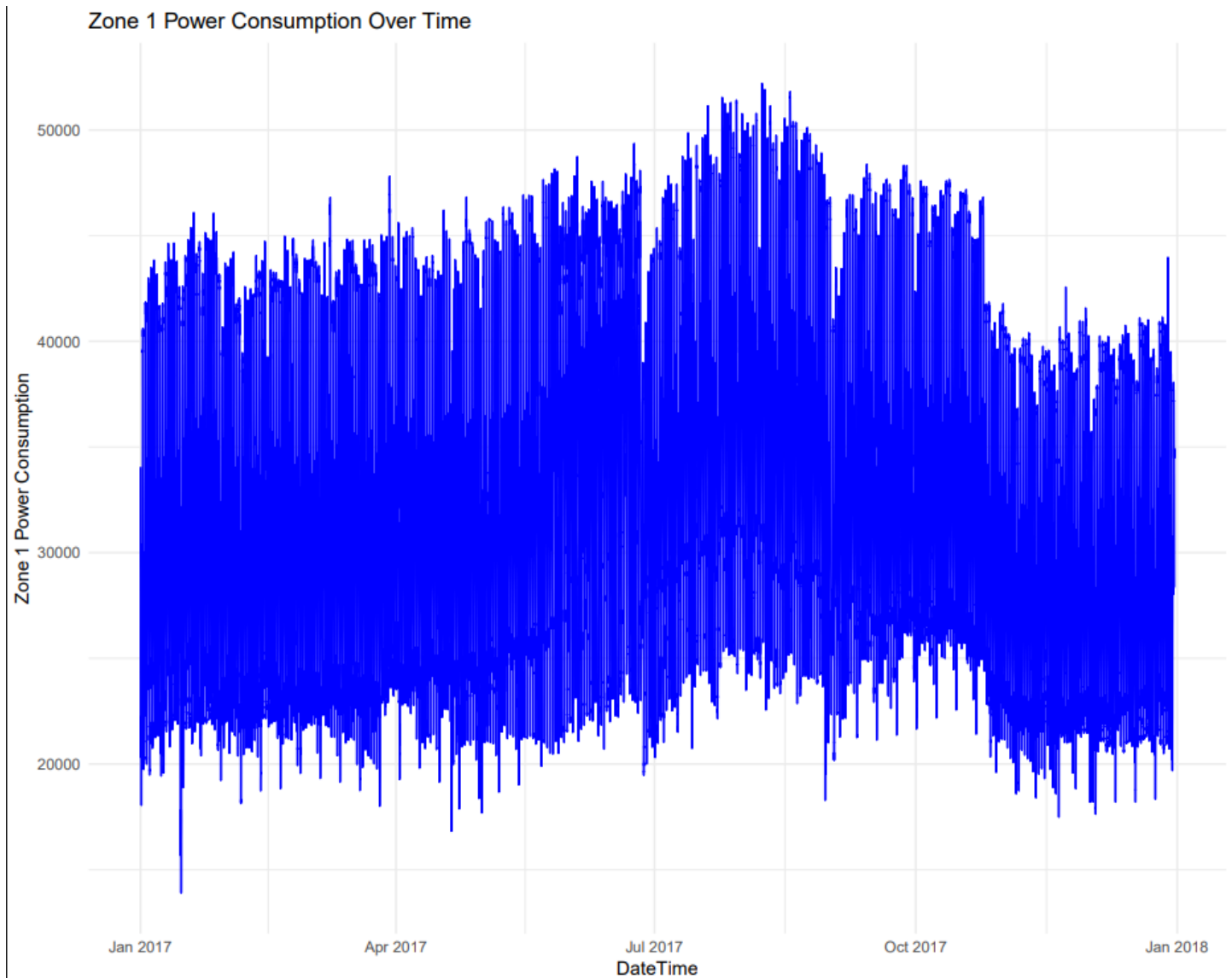


Boxplot: Zone 1 Power



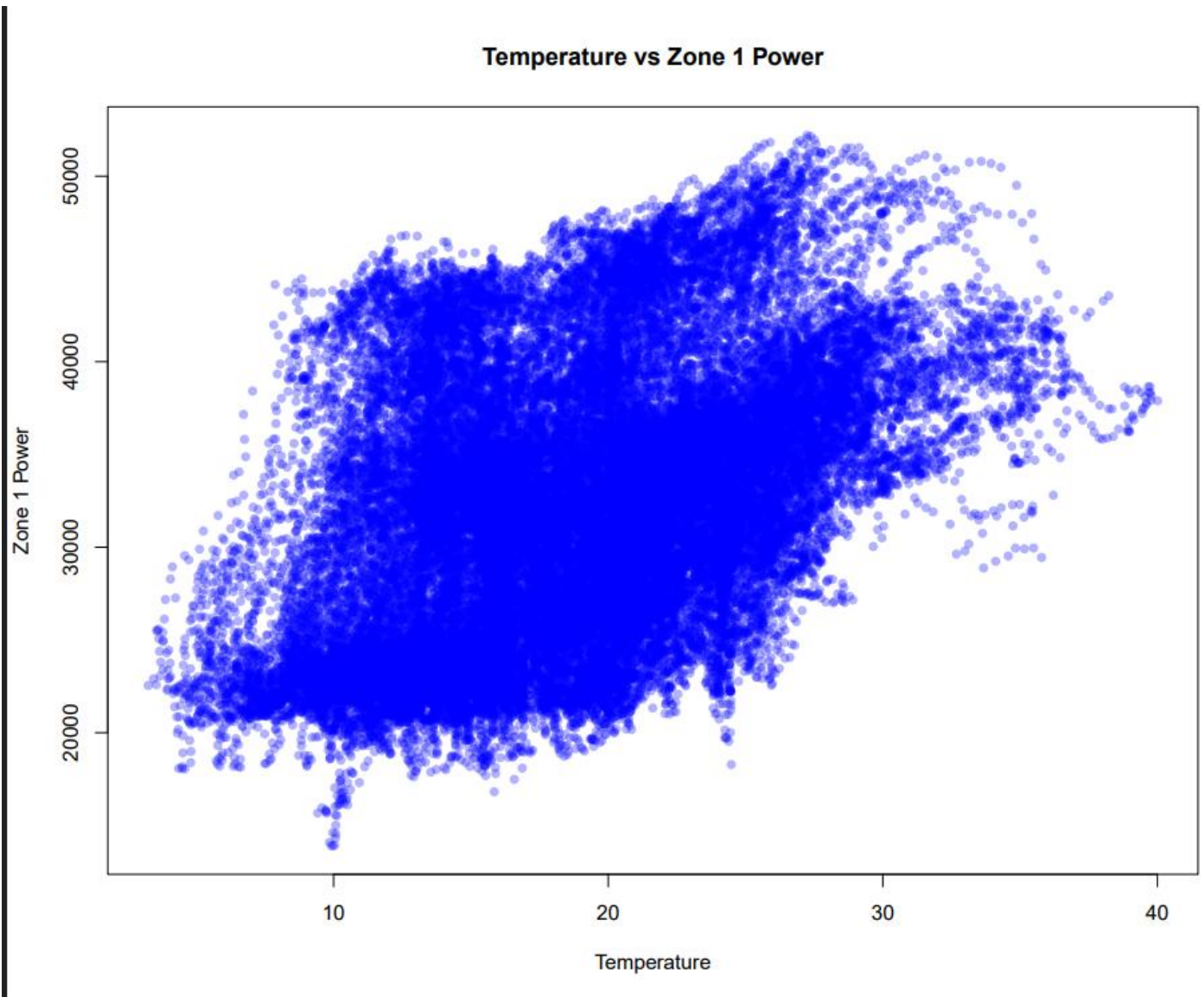
Zone 1 Power Consumption Over Time

The time series plot below shows Zone 1 power consumption throughout 2017. Clear seasonal variation is visible. Power is highest in summer during July – September due to cooling products usage. This confirms that time-based features like Hour and Month can be useful predictors in future models



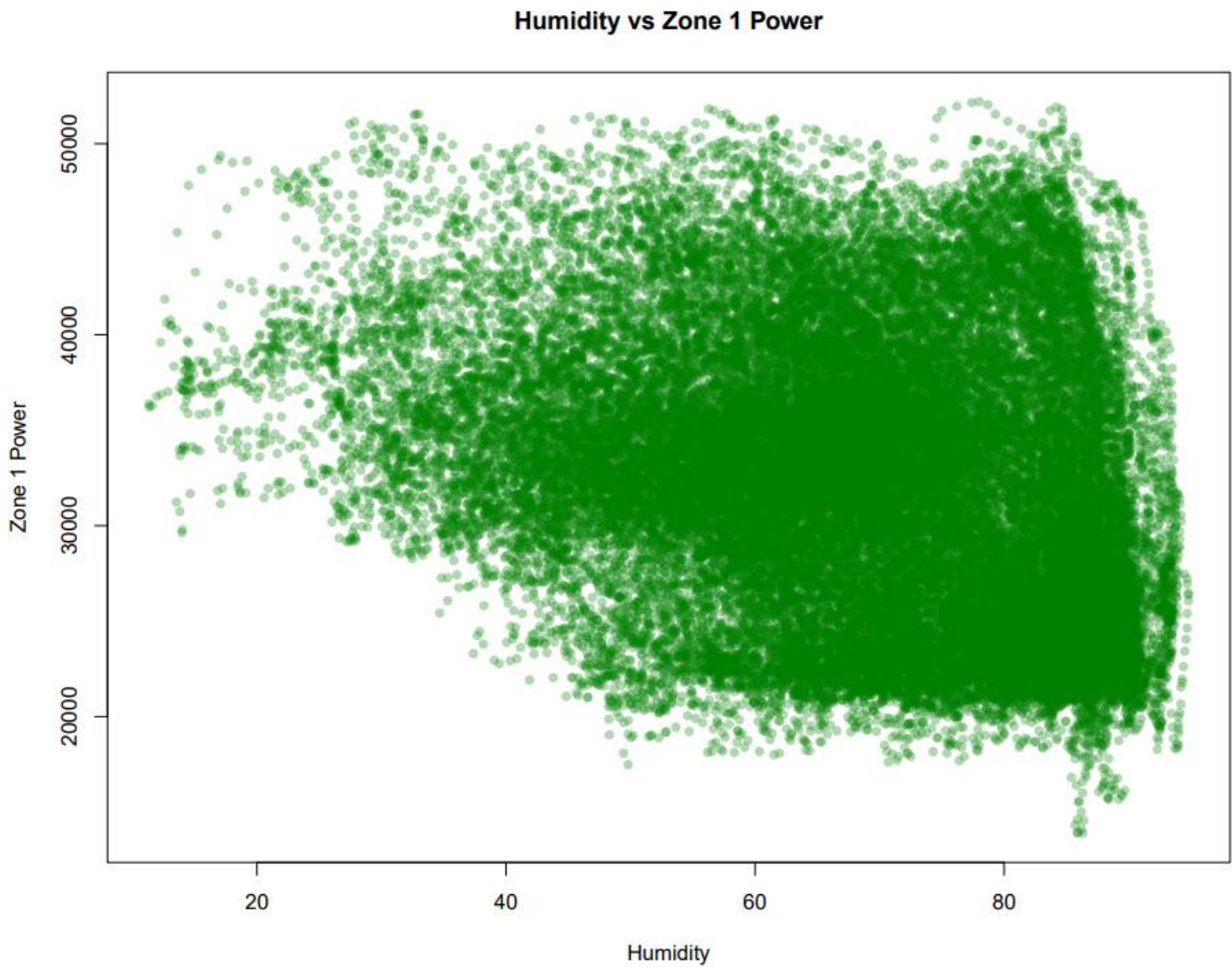
Temperature vs Zone 1 power

A scatter plot of temperature vs Zone 1 power showed nonlinear, U-shaped relationship. Power usage was high at both low temperatures i.e. during winters for heating and at high temperatures i.e. during summers for cooling. It was lower at moderate temperatures. This suggests that temperature is an important predictor, but it may benefit from a squared or polynomial term.



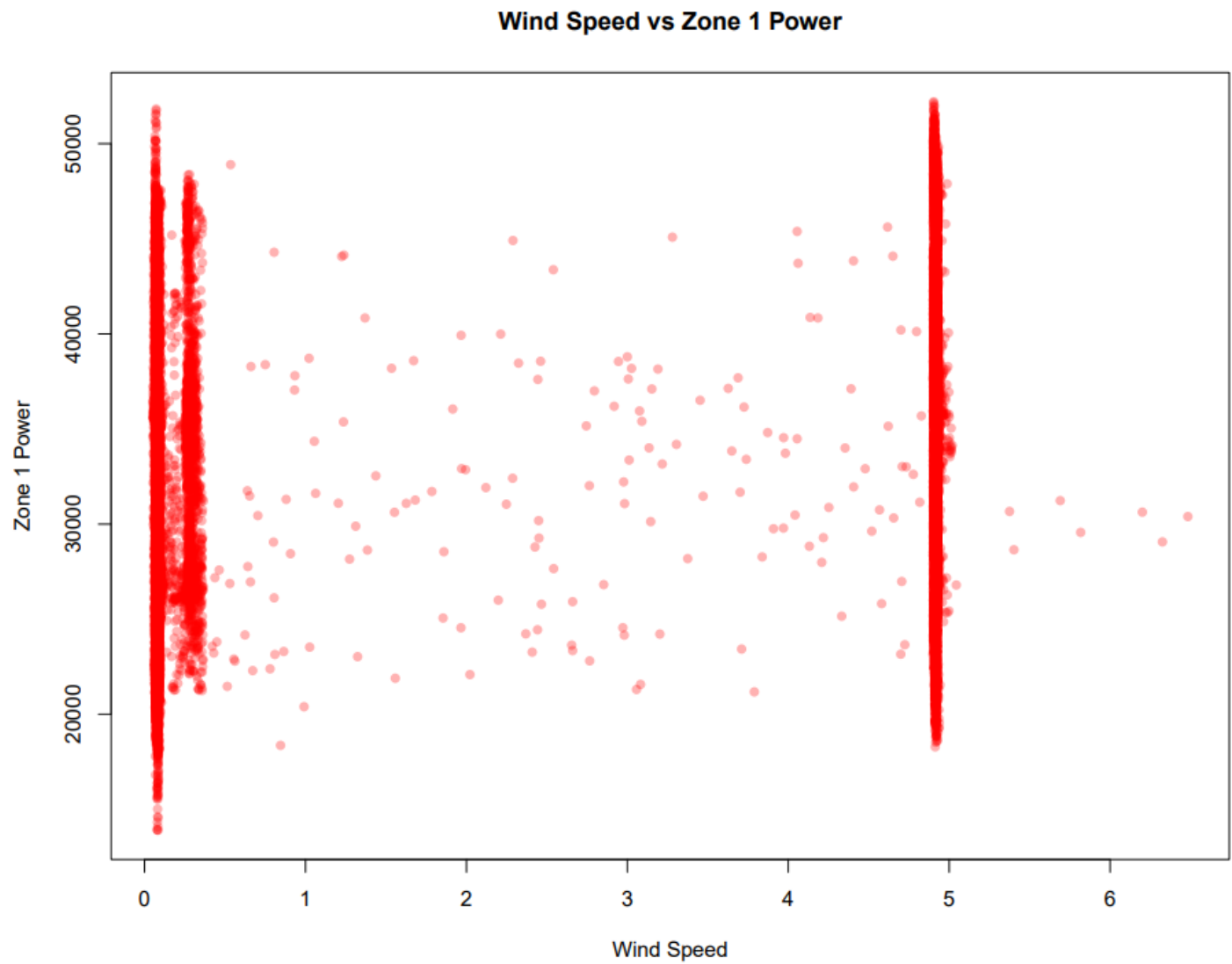
Humidity vs Zone 1 power

Humidity showed moderate negative correlation with the power output. Higher humidity was associated with lower power consumption on average. The relationship was a bit noisy. This suggests that humidity alone may not be a strong predictor



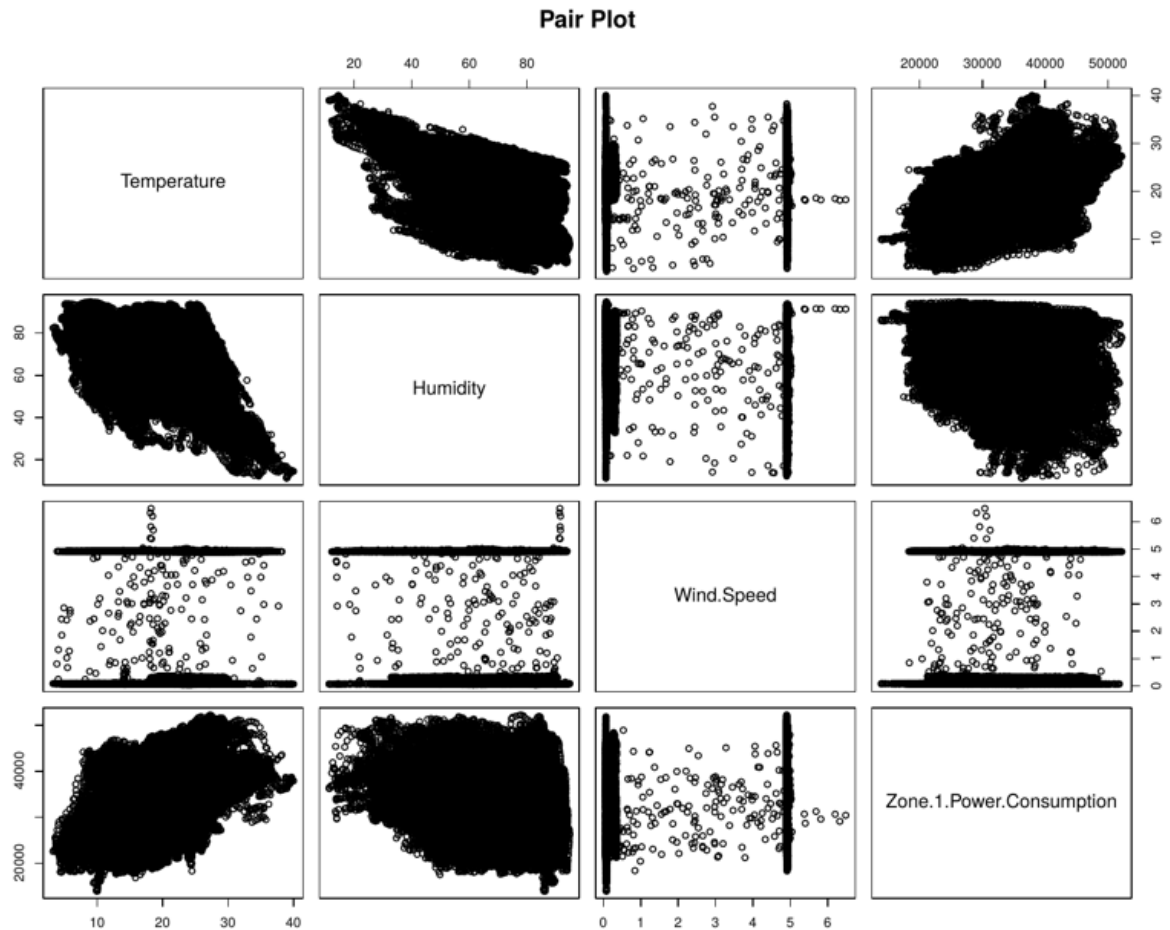
Wind Speed Vs Zone 1 Power

Wind speed data is highly discrete. Most reading falls near zero or around 5 km/h. This produces two dense vertical bands in the scatter plot. There is no clear directional trend between wind speed and Zone 1 power. This suggests that it is a weaker predictor compared to temperature and humidity.



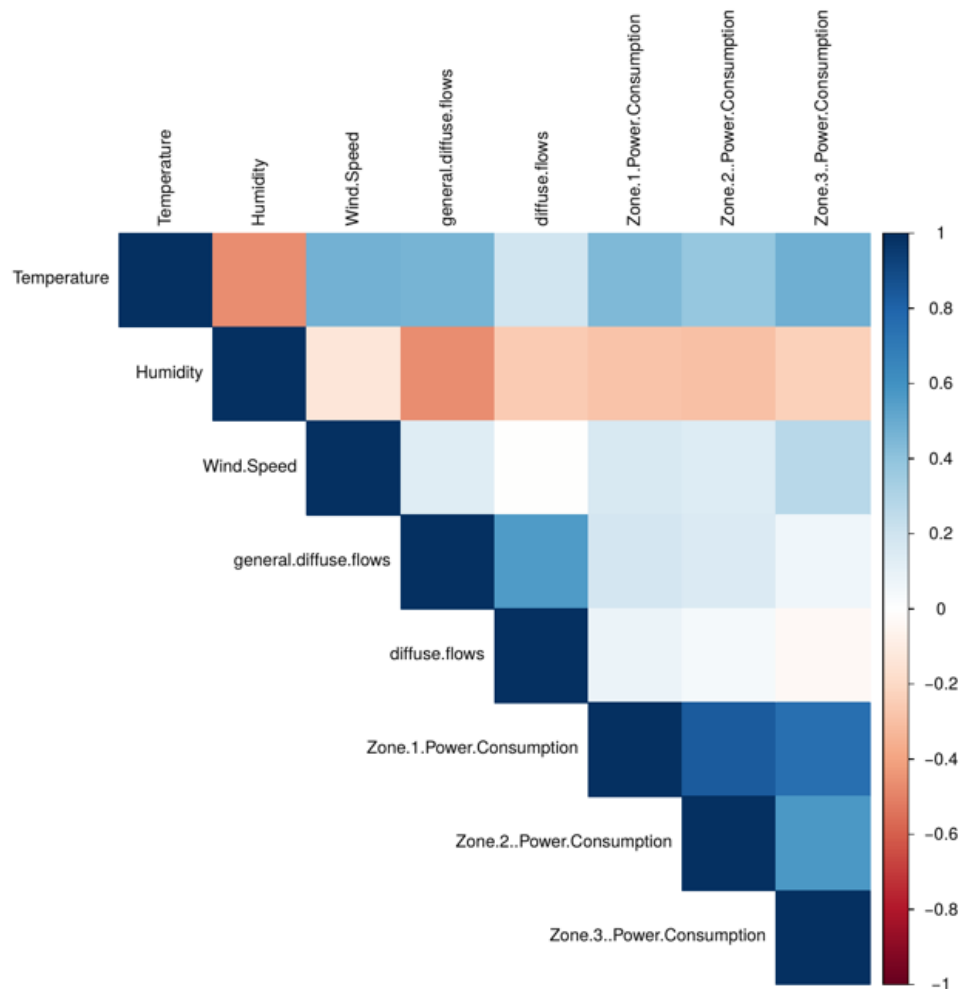
Pair Plots

The pair plot shows all pairwise relationships between predictors and Zone 1 power. It confirms the positive relationship with temperature, negative relationship with humidity and limited signal from wind speed.



Correlation Analysis

Correlation matrix showed that Diffuse Flow and General Diffuse Flows were strongly correlated with each other, which suggested multicollinearity. This motivated us to use LASSO regression, since LASSO can automatically shrink or eliminate redundant predictors. Temperature had moderate positive correlation with power, while humidity showed a negative correlation.



METHODS AND MODELS

We applied three modeling approaches to predict Zone 1 power output. For each method, we describe the model specification, variable selection, tuning, results and interpretation. All models used same training set (80% of data) and evaluated on the same testing set (20% of data). A random seed of `set.seed(2026)` was used. The full set of predictors used across models includes:

Temperature	Humidity
Wind Speed	General Diffuse Flow
Diffuse Flows	Hour of Day
Day of week	Month

Method 1: Multiple Linear Regression (MLR)

Model specification

Multiple Linear Regression models the outcome as a weighted linear combination of predictors:

$$\text{Zone1_Power} = \beta_0 + \beta_1(\text{Temperature}) + \beta_2(\text{Humidity}) + \beta_3(\text{WindSpeed}) + \beta_4(\text{GDF}) + \beta_5(\text{DF}) + \varepsilon$$

Here ε is the random error term which is assumed to follow a normal distribution with mean 0.

Variable Selection and Model Fitting

We first fit a full model with all available predictors. We then examined p-value and variance inflation factors to check multicollinearity. General Diffuse Flows and Diffuse Flows had a variance inflation factor greater than 10. This confirms multicollinearity. We kept both in model since LASSO handles these issues in Method 2. Here we wanted to see the full linear model for comparison

Results

Results produced by Multiple Linear Regression model are as follows:

Predictor	Estimate	Std Error	T-value	p-value
Intercept	26712.35	239.27	111.64	<2e-16
Temperature	535.69	7.17	74.62	<2e-16
Humidity	-55.6957	2.3889	-23.314	<2e-16
Wind Speed	-153.57	15.19	-10.10	<2e-16
General Diffuse Flows	-1.66	0.1639	-10.191	<2e-16
Diffuse Flows	-0.16	0.30	-0.55	0.578

The model achieved an adjusted R^2 of 0.206. All predictors except Diffuse Flows were statistically significant at the 0.05 level.

Interpretation

Each 1°C increase in temperature results with an increase of about 535 MW in Zone 1 power, holding other predictors constant. Higher humidity is associated with lower power use. The General Diffuse Flows and Diffuse Flows coefficient reflect strong daytime activity captures in solar radiation data

Conclusion and Limitation

Linear regression provides solid interpretable baseline. It assumes a linear relationship between predictors and outcomes. This may not fully capture the U-shaped temperature effect or other nonlinear patterns. The test RMSE was approximately 6305.16 MW.

Method 2: Least Absolute Shrinkage & Selection Operator (LASSO) Regression

Model Specification

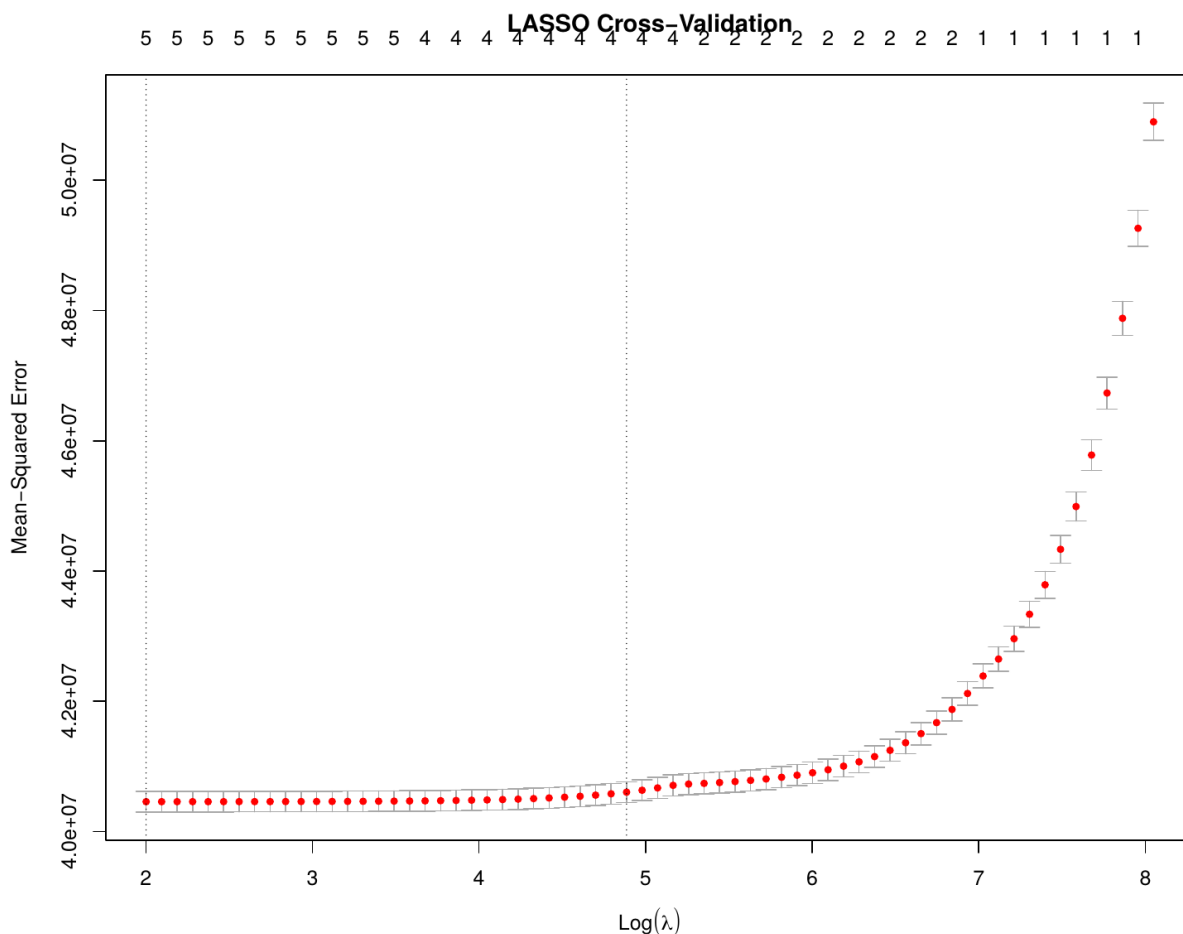
LASSO regression adds penalty on size of the coefficients to standard linear regression loss. The function is:

Minimize: $\sum_i (\text{Zone1_Power}_i - \beta_0 - \beta_1 \text{Temperature}_i - \beta_2 \text{Humidity}_i - \beta_3 \text{WindSpeed}_i - \beta_4 \text{GDF}_i - \beta_5 \text{DF}_i + \lambda(|\beta_1| + |\beta_2| + |\beta_3| + |\beta_4| + |\beta_5|))$

Here lambda is a tuning parameter, and it controls how much shrinkage is applied. When lambda is large, coefficient shrinks towards 0 or are exactly set to 0.

Tuning Parameter Selection

We selected best value of lambda using 10-fold cross validation. The `cv.glmnet()` function in R was used to fit LASSO across the grid of lambda values and compute cross validation mean squared error (CV-MSE) for each. We used `lambda.min` as our final model. This is the lambda value that minimizes CV-MSE. The optimal lambda computed was 7.4028.



Variable Selection Results

After applying LASSO with optimal lambda, all 5 predictors were retained in the model. The coefficients for Diffuse Flows and General Diffuse Flows were slightly shrunk toward each other. This confirms that all predictors carry meaningful information, but LASSO helped reduce the instability caused by their high correlation.

Predictor	LASSO Coefficient
Intercept	26,704.33
Temperature	532.52
Humidity	-55.09
Wind Speed	-146.99
General Diffuse Flows	-1.61
Diffuse Flows	-0.13

In the above table we can see that LASSO model retained all 5 predictors with non-zero coefficients. Temperature shows the largest positive effect. Humidity and Wind Speed are negatively associated with power usage. General Diffuse Flows and Diffuse Flows have nearly zero coefficients.

Results and Interpretations

LASSO model achieved a test RMSE of approximately 6305.17 MW, which is slightly better than linear regression. The coefficient estimates were similar in direction to LOS estimates but with slightly smaller magnitudes due to shrinkage. This confirms the robustness of findings from Method 1 while providing some insurance against overfitting.

Conclusion

LASSO regression performed same as standard linear regression on the test set. It also provides automatic variable selection, which is useful when the number of predictors is large. In our case all predictors survived, which suggests that they are all useful.

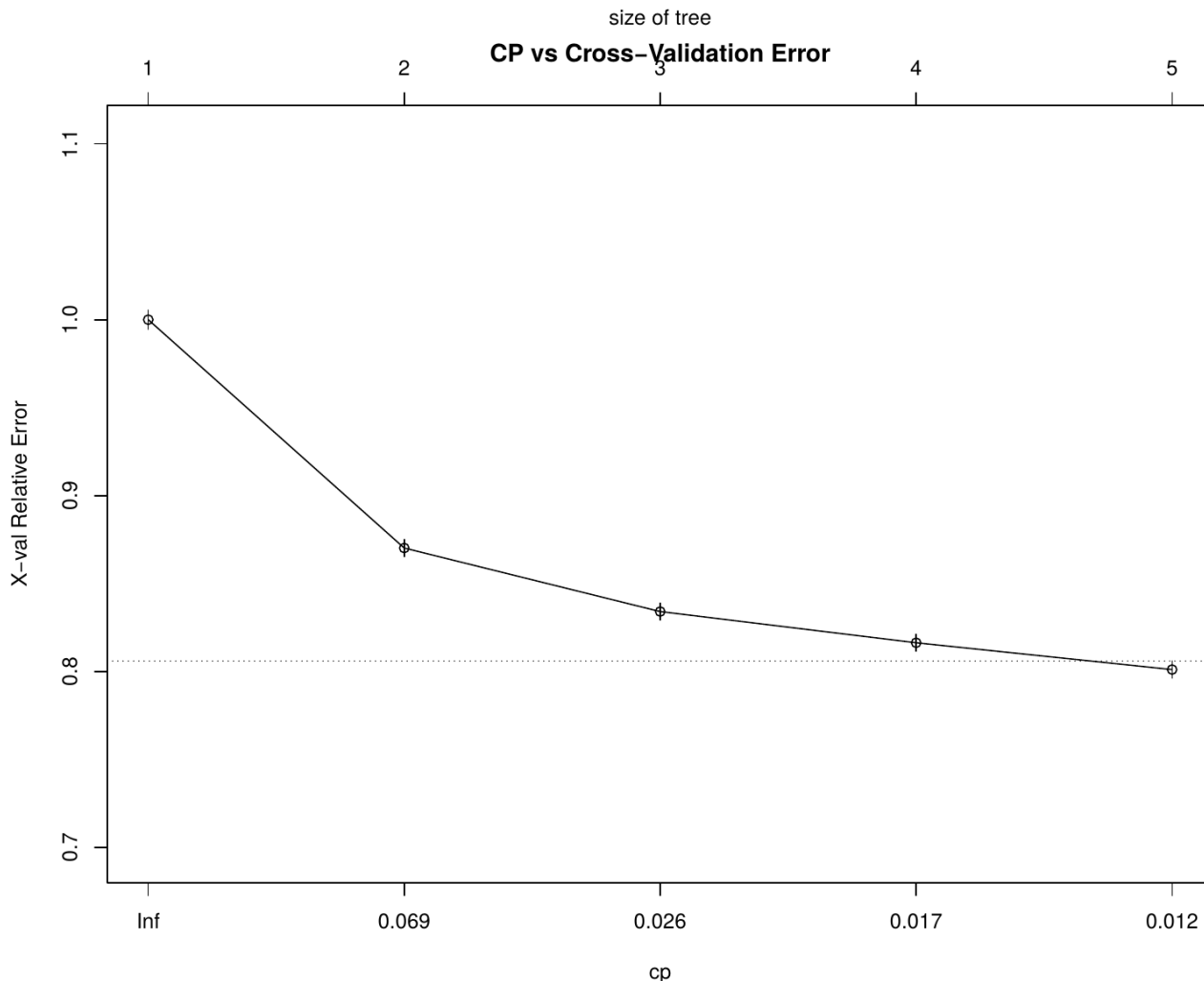
Method 3: Decision Tree

Model Specification

A decision tree is a nonlinear model that recursively splits the data on predictor values to create homogeneous groups. At each split, the algorithm selects the predictor and a split point that most reduces the residual sum of squares. The result is a tree structure where each leaf node predicts the mean outcome value of observations that fall in the group.

Tuning Parameter Selection: Pruning with *cp*

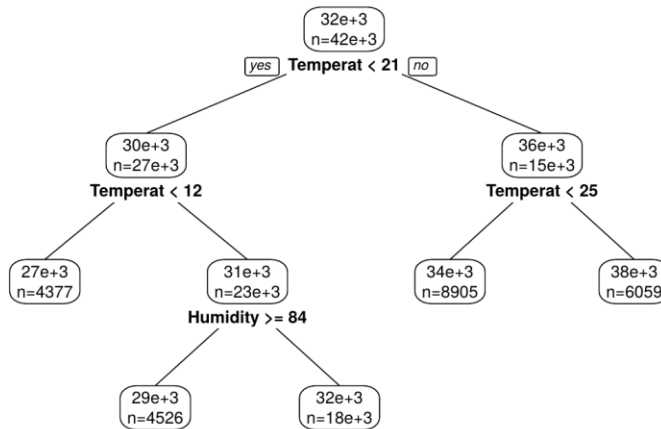
An unpruned tree can grow very deep and overfit the training data. We used cost complexity pruning to find the right tree size. The complexity parameter *cp* controls this. A larger *cp* forces a simpler/smaller tree, whereas small *cp* will allow more splits. We selected optimal *cp* using 10-fold cross validation with *rpart* package in R, choosing the *cp* value that minimized cross validation error. The optimal *cp* was 0.01, which results in a tree with 5 terminal nodes. This value of *cp* represents a deliberate tradeoff. The tree is complex enough and captures the nonlinear patterns in data, but not so deep that it memorizes noise in the training set. The CP table from *rpart* showed 5 candidate *cp* values ranging from 0.130 to 0.010, with cross validation error decreasing from 1.000 to 0.801. Further, reducing *cp* below 0.01 would increase tree complexity with only marginal gains in cross validation errors.



Decision Tree Structure

The pruned tree structure is shown below. On Temperature < 21 C there is root split, which reflects seasonal divide in power usage. Within each branch, further splits on Temperature and humidity improve predictions. The tree has 5 terminal tree nodes. Each node predicts a distinct average power level.

Decision Tree for Power Consumption



Results

The most important splits in the tree were based on Temperature and Humidity only, which captured seasonal and weather driven demand patterns. The decision tree achieved a test RMSE of 6325.27 MW, which is slightly worse than both linear regression approaches.

Interpretation

We found the tree model highly interpretable. For example, when Temperature < 21 C, the tree predicts lower average power around 30,000 MW. When temperature > 21 C and Humidity >= 84%, the tree predicts moderate power. When Temperature ≥ 25°C with low humidity, the model predicts peak power usage around 38,000 MW. These splits align with real world intuition about seasonal electricity demand.

Conclusion

The pruned decision tree captures nonlinear relationships better than linear models and achieved the highest test RMSE among the three methods, performing slightly worse than both linear models. The cp selection via cross validation was essential for finding a tree of appropriate size with 5 terminal nodes.

MODEL COMPARISON and RECOMMENDATION

All 3 models were evaluated on the same 20% test set using RMSE (root mean squared error) and R-squared as performed metrics. RMSE measures the average prediction error in the same units as Zone 1 power, so smaller is better. R-squared measures the proportion of variation explained by model.

Model	Test RMSE(MW)	Test R-squared	Interpretable?	Handles Nonlinearity?
Multiple Linear Regression	6305.16	0.206	Yes	No
LASSO Regression	6305.17	0.206	Yes	No
Decision Tree (Pruned)	6325.27	~0.198	Yes	Yes

Recommendation:

We would recommend the LASSO Regression as the best model for this dataset based on the following reasons:

- LASSO achieved test RMSE of 6305.17 MW, which is identical to MLR. This means that it matches the best prediction accuracy without any loss in performance.
- Unlike standard MLR, LASSO applied regularization which helped address the multicollinearity observed between General Diffuse Flows and Diffuse Flows in the correlation analysis. This makes it more statistical choice.
- All 5 predictors were retained with non-zero coefficients, which confirms that each variable carries meaningful predictive information.
- LASSO remains fully interpretable through its coefficients estimates, just like standard linear regression

Multiple Linear regression is our second recommendation. It achieved same test RMSE of 6305.16 MW and it is the simplest and most transparent model. The Decision Tree, while capturing nonlinear relationships, performed slightly worse with RMSE of 6325.27 MW and as results our third choice.

AREAS OF FUTURE IMPROVEMENT

Based on feedback from professor and our analysis, several improvements could be considered in future work:

- Methods such as Ridge Regression, Random Forest, Gradient Boosting and Best Subset Selection could be compared. Tree methods like Random Forest and Boosting typically outperform simple decision trees by averaging many trees and reducing variance.
- Predictors like Hour, Day of Week and month were extracted during data cleaning but not included in the final models. In future work we can test whether adding these time features improves prediction accuracy or not.
- The dataset contains power data for multiple zones. Including zone specific features or building separate models per Zone could reveal spatial patterns
- Adding a squared temperature term in linear models could better capture the U shaped demand patterns observed in EDA.

CONCLUSION

This project is set out to predict Zone 1 electricity power output using weather and time-based features. Through careful exploratory data analysis, we identify the most important predictors and uncovered key patterns such as time-of-day demand peaks and a nonlinear temperature effect. We applied three modeling approaches: Multiple Linear regression, LASSO Regression, and A Decision Tree and evaluated them on the same test data.

Our main conclusions are:

- MLR and LASSO were the best performing models, both achieved RMSE of ~ 6305 and R squared of 0.206. The Decision Tree performed slightly worse with RMSE of 6325 MW
- LASSO Regression offered slight improvement over standard linear regression and helped address multicollinearity between diffuse flow variables.
- The most important predictors across all models were Humidity, Wind Speed, General Diffuse Flows, Temperature and Diffuse Flows. These variables reflect the daily activity cycle and weather conditions that drive electricity demand.

The models we built practically provide useful predictions and interpretable insights. They could help grid operators anticipate Zone 1 demand under varying weather conditions and plan to supply resources accordingly. Future work should explore methods like Random Forest and Gradient Boosting, which are likely to capture the complex interactions between time, weather and demand even more effectively.